# CS1010 Discrete Mathematics for Computer Science

Shibam Ghosh

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Department of CSE, IIT Hyderabad



# Expected Value and Variance

# Expected Value

## Definition

The **expected value** (mean) of a random variable $X$ on sample space $S$ is:

$$E(X) = \sum_{s \in S} \Pr(s) X(s)$$

## Alternative Definition

If $X$ is a random variable and $\Pr(X = r)$ is the probability that $X = r$, then:

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# Example

**Example:** Suppose in rolling a die, the random variable $X$ denotes the outcome. $X$ takes values $1, 2, 3, 4, 5, 6$ each with probability $1/6$.

$$E(X) = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = \frac{7}{2}$$

**Example:** A ticket costs nothing and pays

- Rs. 100 with probability 0.10,
- Rs. 20 with probability 0.30,
- Rs. 0 with probability 0.60.

The expected winning per ticket: $E(X) = 100(0.10) + 20(0.30) + 0(0.60) = 10 + 6 = 16$.
Thus, over many plays, the **average winning per ticket is Rs. 16**.



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# Interpreting Expected Value

The expected value is a **probability-weighted average** of the possible values of a random variable.

$$E(X) = \sum_{x} x \Pr(X = x).$$

It does **not** necessarily mean the value that we will actually observe. For example, when rolling a fair die,

$$E(X) = \frac{7}{2} = 3.5.$$

But a single roll can never produce 3.5. Instead, the expected value describes the **long-run average** over many repetitions.



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# Linearity of Expected Value

**Theorem**

If $X_i$ are random variables on $S$, and $a, b \in \mathbb{R}$:

(i) $E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$

(ii) $E(aX + b) = aE(X) + b$

**Proof:** Exercise. **Example:** A fair coin is tossed 100 times. Let

$$X_i = \begin{cases} 1, & \text{if the } i\text{-th toss is Head,} \\ 0, & \text{otherwise.} \end{cases}$$

Then the total number of heads is $X = X_1 + X_2 + \cdots + X_{100}$. Since $E(X_i) = \frac{1}{2}$, by linearity of expectation, $E(X) = 100 \times \frac{1}{2} = 50$.

Thus, on average, we expect **50 heads** in 100 tosses.



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# Application of Linearity of Expectation

Suppose $n$ people randomly receive $n$ hats, one hat per person. Let $X$ be the number of people who receive their own hat. For each person $i$, define

$$X_i = \begin{cases} 1, & \text{if person } i \text{ gets their own hat,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = X_1 + \cdots + X_n.$$

For each $i$,

$$\Pr(X_i = 1) = \frac{1}{n}, \qquad E(X_i) = \frac{1}{n}.$$

Therefore, by linearity,

$$E(X) = \sum_{i=1}^{n} E(X_i) = n \cdot \frac{1}{n} = \boxed{1}.$$

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# Expected Value of a Binomial Random Variable

**Theorem**

If $X$ denotes the number of successes in $n$ Bernoulli trials, each having success probability $p$, then $E(X) = np$.

**Proof:** For each trial $i = 1, 2, \ldots, n$, define the indicator random variable

$$X_i = \begin{cases} 1, & \text{if the } i\text{-th trial is a success,} \\ 0, & \text{otherwise.} \end{cases} \quad \text{and } X = X_1 + X_2 + \cdots + X_n.$$

Since $\Pr(X_i = 1) = p, \Pr(X_i = 0) = 1 - p$, we have $E(X_i) = 0(1-p) + 1(p) = p$.

Using the linearity of expectation,

$$\begin{aligned} E(X) &= E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n) \\ &= p + p + \cdots + p = np. \end{aligned}$$

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# Why Is This Example Interesting?

The events

$$X_1 = 1,\ X_2 = 1, \ldots, X_n = 1$$

are **not independent**.

Nevertheless,

$$E(X_1 + \cdots + X_n) = E(X_1) + \cdots + E(X_n).$$

**Linearity of expectation does not require independence.**



# Independent Random Variables

**Definition**

Random variables $X$ and $Y$ are independent if:

$$\Pr(X = r_1 \text{ and } Y = r_2) = \Pr(X = r_1) \cdot \Pr(Y = r_2)$$

$\mathcal{Q}$. Show that, if $X$ and $Y$ are independent, then $E(XY) = E(X)E(Y)$.

$\mathcal{A}$. Exercise.

# Expected Value of a Function

If $X$ is a discrete random variable and $g$ is a function, then

$$\boxed{E[g(X)] = \sum_x g(x) \Pr(X = x)}$$

For example,

$$E(X^2) = \sum_x x^2 \Pr(X = x).$$

Another example, consider the linear function $g(x) = ax + b$,

$$E[aX + b] = aE[X] + b.$$

This idea will be useful when we study **variance**.

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# Expectation of a Geometric Random Variable

Let $X$ be a geometric random variable with success prob. $p$, and let $q = 1 - p$. Then

$$\Pr(X = k) = q^{k-1}p, \qquad k = 1, 2, \ldots \implies E[X] = p\sum_{k=1}^{\infty} kq^{k-1}.$$

- Let $S = \sum_{k=1}^{\infty} kq^{k-1}$. Then

$$S = 1 + 2q + 3q^2 + 4q^3 + \cdots,$$
$$qS = q + 2q^2 + 3q^3 + 4q^4 + \cdots.$$

- Subtracting, $S - qS = 1 + q + q^2 + q^3 + \cdots = \frac{1}{1-q} \implies S = \frac{1}{(1-q)^2} = \frac{1}{p^2}$.
- Hence

$$E[X] = pS = \frac{1}{p}.$$

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$$E[X] = pS = \frac{1}{p}.$$

# Expectation of a Poisson Random Variable

Let $X \sim \text{Poisson}(\lambda)$. Then

$$\Pr(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, k = 0, 1, 2, \ldots \implies E[X] = \sum_{k=0}^{\infty} k\, e^{-\lambda}\frac{\lambda^k}{k!}.$$

The $k = 0$ term is zero, so

$$E[X] = e^{-\lambda}\sum_{k=1}^{\infty} k\frac{\lambda^k}{k!}. = e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^k}{(k-1)!} = \lambda e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!}.$$

Letting $j = k - 1$,

$$E[X] = \lambda e^{-\lambda}\sum_{j=0}^{\infty}\frac{\lambda^j}{j!} = \lambda e^{-\lambda}e^{\lambda} = \lambda.$$

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# Interpreting the Poisson Parameter

For $X \sim \text{Poisson}(\lambda)$, we have $\boxed{E[X] = \lambda}$.

Thus, $\lambda$ represents the expected number of events in the interval being considered.

For example, if the number of requests received by a server in one second follows

$$X \sim \text{Poisson}(5),$$

then

$$E[X] = 5.$$

So we expect, on average, 5 requests per second.



## Expected Value and Variance

# Variance

# Motivation: Expected Value Is Not the Whole Story

Suppose you are playing two games:

| | |
|---|---|
| Game A | Always receive 50 points. |
| Game B | Receive 0 points with probability $1/2$,<br>Receive 100 points with probability $1/2$. |

Both have the same expected value: $E(A) = 50, E(B) = 0 \cdot \frac{1}{2} + 100 \cdot \frac{1}{2} = 50$.

However, the two games are very different.

- Game A always gives exactly 50 points.
- Game B is highly unpredictable.

**How can we measure this uncertainty or spread?**

The **variance** measures how much the values of a random variable tend to **deviate** from its **expected** value.

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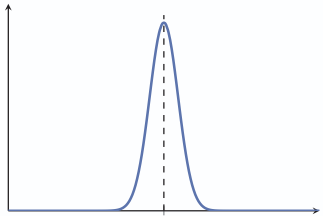
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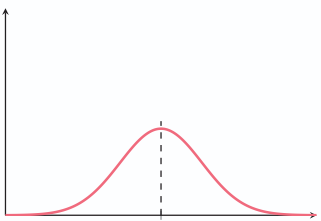
# Motivation: Same Mean, Different Spread

Small Variance




$E(X) = 5$

Large Variance




$E(X) = 5$

Both random variables have the **same expected value**, but the one on the right is much more **spread out**: **Variance measures this spread.**

# Variance

## Definition

The **variance** $V(X)$ of a random variable $X$ is:

$$V(X) = E\left[(X - E[X])^2\right] = \sum_{s \in S} (X(s) - E(X))^2 \Pr(s) = E((X - E(X))^2)$$

- The **standard deviation** $\sigma(X) = \sqrt{V(X)}$.
- $V(X) = E(X^2) - E(X)^2$

$\mathcal{Q}$. What is the variance of a fair die roll?

$\mathcal{A}$. We know $E(X) = 7/2$. Compute $E(X^2)$:

$$E(X^2) = \frac{1}{6}(1^2+2^2+3^2+4^2+5^2+6^2) = \frac{91}{6}. \text{ So, } V(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

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# An Alternative Formula for Variance

**Theorem**

For any random variable $X$, $V(X) = E(X^2) - [E(X)]^2$

**Proof.**

- Let $\mu = E(X)$. From the definition of variance,

$$V(X) = E\left[(X-\mu)^2\right] = E\left[X^2 - 2\mu X + \mu^2\right] = E(X^2) - 2\mu E(X) + E(\mu^2).$$

- Since $\mu$ is a constant, $E(\mu^2) = \mu^2$.
- Therefore,

$$V(X) = E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2 = E(X^2) - [E(X)]^2.$$



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# Properties of Variance

- Since $(X - E(X))^2 \geq 0$, we always have

$$\boxed{V(X) \geq 0}.$$

- Moreover, $V(X) = 0$ if and only if $X$ always takes the same value. For example, if

$$X = 5$$

with probability 1, then

$$E(X) = 5, \qquad V(X) = 0.$$

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# Bienaymé's Formula

**Theorem**

If $X$ and $Y$ are **independent** random variables: $V(X+Y) = V(X) + V(Y)$

**Proof.** Using $V(Z) = E(Z^2) - E(Z)^2$, we have

$$\begin{aligned} V(X+Y) &= E((X+Y)^2) - E(X+Y)^2 \\ &= E(X^2 + 2XY + Y^2) - (E(X) + E(Y))^2 \\ &= E(X^2) + 2E(XY) + E(Y^2) \\ &\quad - E(X)^2 - 2E(X)E(Y) - E(Y)^2. \end{aligned}$$

Since $X$ and $Y$ are independent, $E(XY) = E(X)E(Y)$. Therefore, the cross terms cancel:

$$V(X+Y) = [E(X^2) - E(X)^2] + [E(Y^2) - E(Y)^2] = V(X+Y) = V(X) + V(Y).$$

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# Bienaymé's Formula: General Case

If $X_1, \ldots, X_n$ are **pairwise independent** random variables, then

$$\boxed{V\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} V(X_i)}$$

- For independent Bernoulli random variables with success probability $p$, we have $E(X_i) = p$ and $E(X_i^2) = p$, because $X_i^2 = X_i$. Thus,

$$V(X_i) = E(X_i^2) - E(X_i)^2 = p - p^2 = p(1 - P)$$

- If $X = X_1 + \cdots + X_n$ is the number of successes, then

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# Bienaymé's Formula: General Case

If $X_1, \ldots, X_n$ are **pairwise independent** random variables, then

$$\boxed{V\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} V(X_i)}$$

- For independent Bernoulli random variables with success probability $p$, we have $E(X_i) = p$ and $E(X_i^2) = p$, because $X_i^2 = X_i$. Thus,

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# Variance Under Scaling and Translation

$\mathcal{Q}$. For constants $a, b \in \mathbb{R}$, show that $V(aX + b) = a^2 V(X)$.

$\mathcal{A}$. Proof. Let $Y = aX + b$. First, by linearity of expectation, $E(Y) = E(aX + b) = aE(X) + b$.

- Therefore, from the definition of variance,

$$V(Y) = E\left[(Y - E(Y))^2\right] = E\left[(aX + b - (aE(X) + b))^2\right] = E\left[(a(X - E(X)))^2\right]$$
$$= E\left[a^2(X - E(X))^2\right] = a^2 E\left[(X - E(X))^2\right] = a^2 V(X).$$

**Interpretation:**

- Adding a constant shifts all outcomes but does not change their spread.
- Multiplying all outcomes by $a$ multiplies the spread by $|a|$, and therefore the variance by $a^2$.



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## Expected Value and Variance

# Probabilistic Bounds

# Why Do We Need Probability Bounds?

Suppose $X$ is a random variable describing some quantity of interest. Often, we would like to answer questions such as

$$\Pr(X \geq a) \qquad \text{or} \qquad \Pr(|X - \mu| \geq k).$$

1. If we know the **exact distribution** of $X$, we can compute these probabilities.
2. In many applications, however, the exact distribution is unknown.
3. We may nevertheless know some basic information, such as $E[X]$ and $V(X)$.

**Markov's and Chebyshev's inequalities** provide useful upper and lower bounds using only limited information about $X$.



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# Markov's Inequality

Let $X$ be a non-negative random variable. Then for any $a > 0$,

$$\boxed{\Pr(X \geq a) \leq \frac{E[X]}{a}}.$$

**Proof.**

$$E[X] = \sum_x x \Pr(X = x) \geq \sum_{x \geq a} a \Pr(X = x) = a \Pr(X \geq a) \implies \Pr(X \geq a) \leq \frac{E[X]}{a}.$$



# Chebyshev's Inequality

Let $X$ be a random variable with $E[X] = \mu$, $V(X) = \sigma^2$. Then, for any $k > 0$,

$$\Pr(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2} \quad \text{or equivalently,} \quad \Pr(|X - \mu| < k) \geq 1 - \frac{\sigma^2}{k^2}.$$

- Apply Markov's inequality to the non-negative random variable $Y = (X - \mu)^2$.
- For $k > 0$,

$$\Pr(|X - \mu| \geq k) = \Pr\left((X - \mu)^2 \geq k^2\right).$$

- By Markov's inequality,

$$\Pr\left((X - \mu)^2 \geq k^2\right) \leq \frac{E[(X - \mu)^2]}{k^2} = \frac{V[X]}{k^2} = \frac{\sigma^2}{k^2}.$$

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# Application of Chebyshev's Inequality

Suppose the number of items produced in a factory during a week is a random variable $X$ with

$$E[X] = 500, \qquad \mathrm{Var}(X) = 100.$$

$\mathcal{Q}$. What can be said about $\Pr(X \geq 700)$?

$\mathcal{A}$. By Markov's inequality, $\Pr(X \geq 700) \leq \frac{500}{700} = \frac{5}{7}$.

$\mathcal{Q}$. What can be said about $\Pr(400 \leq X \leq 600)$?

$\mathcal{A}$. Since $|X - 500| \geq 100$ is the complement, Chebyshev gives
$\Pr(|X - 500| \geq 100) \leq \frac{100}{100^2} = \frac{1}{100}$.
Hence, $\Pr(400 \leq X \leq 600) \geq \frac{99}{100}$.

# Questions and Answers on Expected Value and Variance

# Expected Number of Die Rolls

Suppose that we roll a fair die until a 6 comes up or we have rolled it 10 times. What is the expected number of times we roll the die?

Let $X$ be the number of rolls.

- For $k = 1, \ldots, 9$, we stop at roll $k$ if the first $k-1$ rolls are not 6, followed by a 6: $\Pr(X = k) = \left(\frac{5}{6}\right)^{k-1} \frac{1}{6}$.
- We roll exactly 10 times if no 6 occurs in the first 9 rolls: $\Pr(X = 10) = \left(\frac{5}{6}\right)^{9}$.
- Therefore,

$$E[X] = \sum_{k=1}^{9} k \left(\frac{5}{6}\right)^{k-1} \frac{1}{6} + 10 \left(\frac{5}{6}\right)^{9}$$



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# Expected Number of Trials to Find a Prime

Estimate the expected number of integers with 1000 digits that need to be selected at random to find a prime, if the probability that a 1000-digit integer is prime is approximately $\frac{1}{2302}$.

- Let $X$ be the number of integers selected until the first prime is found.
- Therefore, $X$ follows a geometric distribution with success probability $p = \frac{1}{2302}$.
- The expected value of a geometric random variable is

$$E[X] = \frac{1}{p} = 2302.$$



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# Variance of the Number of Heads

What is the variance of the number of heads that come up when a fair coin is flipped 10 times?

Let $X$ be the number of heads obtained in 10 flips.

- Each flip is a Bernoulli trial with

$$p = \Pr(\text{Head}) = \frac{1}{2}.$$

- Therefore,

$$X \sim \text{Binomial}\left(10, \frac{1}{2}\right).$$

- For a binomial random variable,

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# Application of Chebyshev's Inequality

The lifetime $X$ (in hours) of a certain type of battery satisfies $E[X] = 100$, $V[X] = 400$. No assumption is made about the distribution of $X$. How large must $k$ be to guarantee that $\Pr(100 - k < X < 100 + k) \geq 0.95$?

$\mathcal{A}$. By Chebyshev's inequality,

$$\Pr(|X - 100| \geq k) \leq \frac{400}{k^2} \implies \Pr(|X - 100| < k) \geq 1 - \frac{400}{k^2}.$$

$\mathcal{A}$. We want $1 - \frac{400}{k^2} \geq 0.95$.

- Hence,

$$\frac{400}{k^2} \leq 0.05 \implies k^2 \geq 8000 \implies k \geq 40\sqrt{5} \approx 89.44.$$

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# Thank You for your attention!

Any questions?